# Distributed Computing with Spark for Actionable Business Insights!

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SPARK SUMMIT EUROPE 2016

# Who I am

- Stephan Kessler
- SAP HANA Vora Team, Walldorf, Germany
- Integrating SAP engines into Apache Spark since almost two years
- 2nd Spark Summit as a speaker

# Today's talk

> On average, between 60% and 73% of all data within an enterprise goes unused for business intelligence (BI) and analytics.

The Forrester Wave ™: Big Data Hadoop Distributions, Q1 2016," January 19, 2016 by Mike Gualtieri and Noel Yuhanna

> Skills gaps continue to be a major adoption inhibitor for 57% of respondents, while deciding how to get value from Hadoop was cited by 49% of respondents.

Gartner: "Survey Analysis: Hadoop Adoption Drivers and Challenges," May 12, 2015 by Nick Heudecker and Merv Adrian

# Current System Landscape

# What is missing?

- Business application perspective:
  - Access to Big Data Landscape in a standardized way
  - Similar SQL expressiveness
- Big Data / Data Science perspective:
  - Access to specialized engines to perform analysis close to the data
  - Integration of ‘business engines’ into Spark

# SAP Hana Vora

# SAP Hana Vora – 10k ft POV

Data Science, Predictive, Business Intelligence, Visualization Apps

## SAP HANA Vora

Data Modeler

| OLAP | Time Series | Graph | Doc Store |
|---|---|---|---|

| Disk-to-Memory Accelerator | Distributed Transaction Log |
|---|---|

## ERP Systems

SAP HANA Platform

Spark

hadoop

# Goals

- Make data and functionality available in enterprise applications as well as Spark applications
- Allow an easy consumption, i.e., allow users to write SQL for computation jobs

# Agenda

- Business functionality integration in Spark
- Utilizing different data sources in Spark
- HANA Vora 1.3
- Summary & Outlook

# Focus on Spark User POV

Data Science, Predictive, Business Intelligence, Visualization Apps

## SAP HANA Vora

Data Modeler

OLAP

Time Series

Graph

Doc Store

Disk-to-Memory Accelerator

Distributed Transaction Log

## ERP Systems

SAP HANA Platform

Spark

hadoop

# Business Functionality in Spark

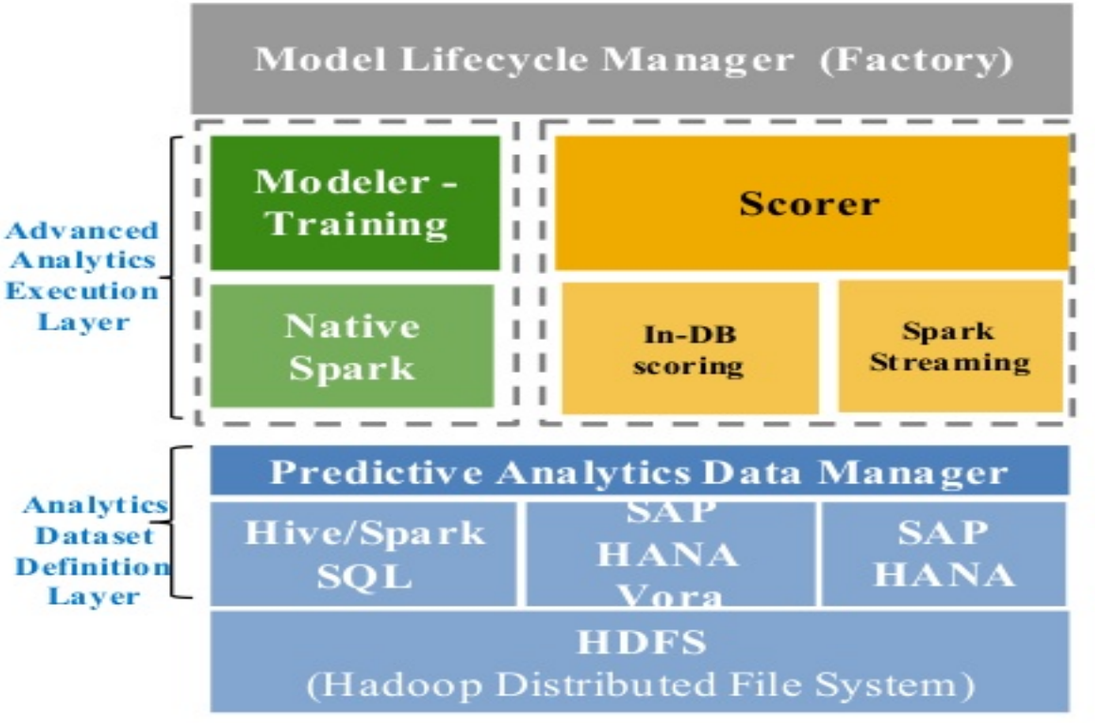
- Questions answered in Spark:
  - What is the sentiment of users for product XYZ?
  - When will a certain part of this machinery fail?
- You might also want to know…
  - What is the sales volume for XYZ?
  - How much did this part cost?

# Example: SAP Predictive Analytics and SAP HANA Vora

1. Business Analyst friendly – No coding or in-depth Big Data expertise required
2. Support for end-to-end operationalisation of predictive models on Hadoop
   - Data preparation of Analytical Dataset for modelling
   - Native Spark Modelling – Ultra wide datasets
   - Scoring using In Database Apply or Spark Stream API

- **Usage of different sources (Vora, Hana, Spark, …)**

http://go.sap.com/solution/platform-technology/analytics/predictive-analytics.html



# Business Functionality in Spark

- Important typical ERP function
  - Currency Conversion (i.e., EUR → GBP)
  - Done via SQL UDF
- Required to analyze enterprise data in Spark

# Currency Conversion

| TID | USERID | CURRENCY | AMOUNT | ORDERDATE |
|---|---|---|---|---|
| 100 | User1 | USD | 120.10 | 2014-12-15 |
| 101 | User1 | USD | 24.99 | 2015-01-01 |
| 102 | User5 | EUR | 24.11 | 2015-01-02 |
| 103 | User3 | DBP | 542.00 | 2015-01-02 |

- Single currency makes transactions comparable
- Conversion not trivial: rates change over time

# Currency Conversion

- Introducing an UDF implemented in Spark

```
CC( AMOUNT Double,
        SOURCE_CURRENCY String,
        TARGET_CURRENCY String,
        REF_DATE  String )
```

- Converting everything in USD

```
SELECT TID, USERID, ORDERDATE,
        CC( AMOUNT, CURRENCY, "USD", ORDERDATE )
FROM ORDERS
```

# Currency Conversion

- Conversion backed by a ‘rates’ table

| SOURCE_CUR | TARGET_CUR | REF_DATE | RATE |
|---|---|---|---|
| EUR | USD | 2015-01-01 | 1.32113 |
| EURO | USD | 2015-01-02 | 1.30121 |
| USD | GBP | 2015-01-01 | 0.68960 |

- Calculation simple, maintenance difficult
- Rates maintained in ERP system
  - Couldn’t we use that?

# Specialized Engine: Time Series

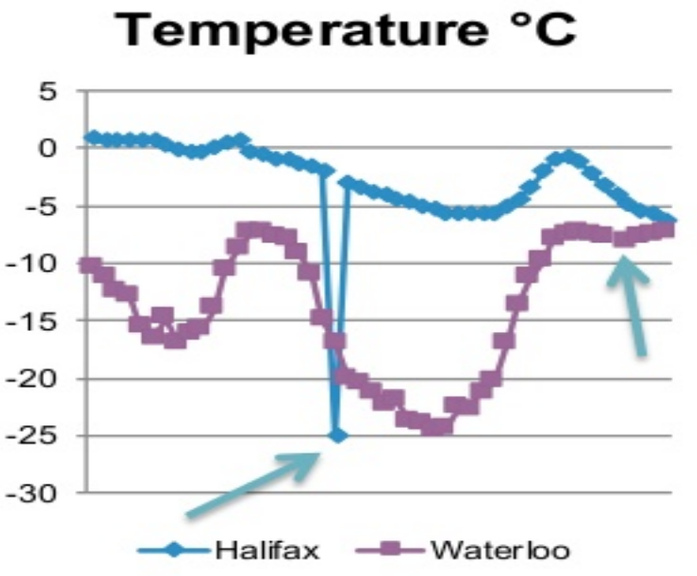
- HANA Vora Time Series Engine in a nutshell:
  - Effective Model-Based Compression
  - Multi-representation storage for time series
- Optimized usage for IOT applications
  - Fast injections paths
  - Long running processes in a cluster

# Specialized Engine: Time Series

- Query Language: SQL Dialect
- How to use that in Spark?

```
SELECT STDDEV(val1)
FROM SERIES ts
BETWEEN "2000-01-01", "2001-12-31"

SELECT TREND(val1) OVER (SERIES)
FROM SERIES ts
BETWEEN "2000-01-01", "2001-12-31"
```



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# What we have seen so far

- Currency Conversion:
  - Implementation in Spark SQL
  - Computation could be deferred

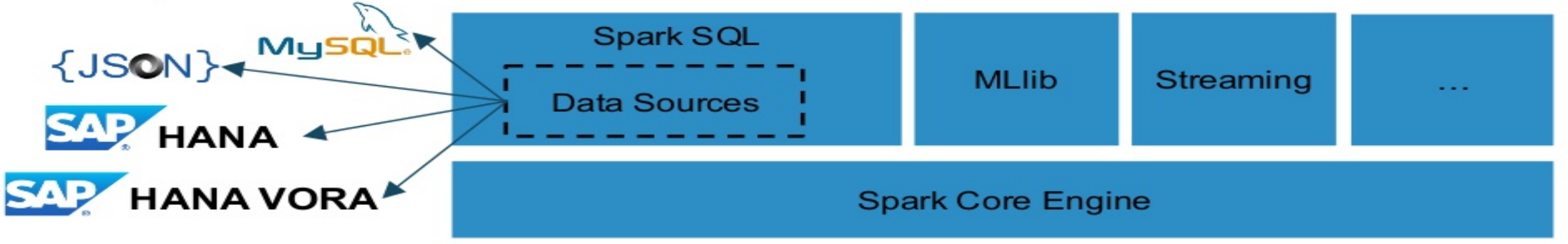
→ "The pushdown of everything"

- Time Series Engine:
  - Special query language
  - .. but no implementation in Spark

→ Raw SQL

# The Pushdown of Everything

- Spark datasource API




- Limited to *Filters* and *Projects*

# Vora Extension to Datasource API

- Datasource indicates its processing capabilities
- Arbitrary parts of logical plan can be computed where the data is
- Details in Spark Summit Europe Talk 2015
  - https://www.youtube.com/watch?v=QNaf2Z8I8IY
  - *"The Pushdown of Everything"*

# Vora Extension to Datasource API

- Consider this query

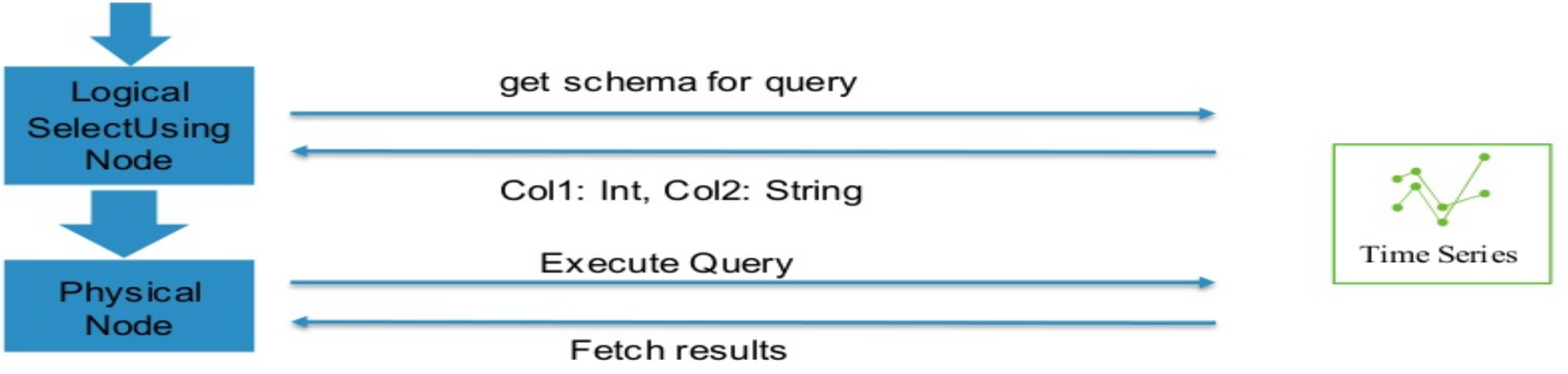
```
SELECT ORDERDATE,
        AVG(CC( AMOUNT, CURRENCY, "USD", ORDERDATE ))
FROM ORDERS
GROUP BY ORDERDATE
```

- Pushing down filters an projects: *SCAN on orders*
- Pushing down arbitrary parts returns:
  - One row per orderdate
  - Converted currency

# Raw SQL Extension

- Query Syntax on SparkSQL not supported but in the datasource → Raw SQL

```
``select trend(val1) from ts …`` using com.sap.spark.engines
```



# Pushdown & Raw SQL

- Both extensions allow to incorporate other data sources extensively
- Computation happens where the data is
- Integration is mostly seamless for Spark developer
- Interfaces are open source:
  - https://github.com/SAP/HANAVora-Extensions

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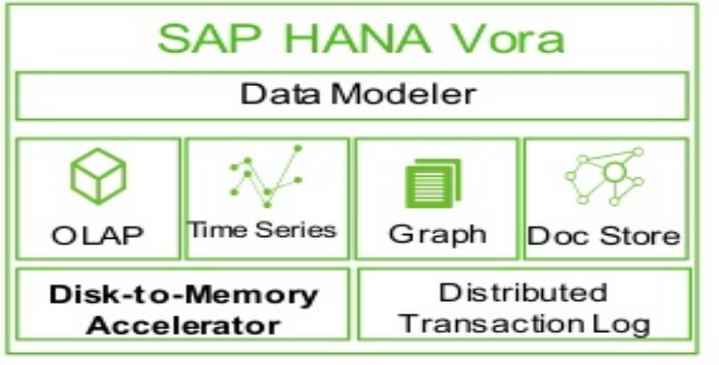
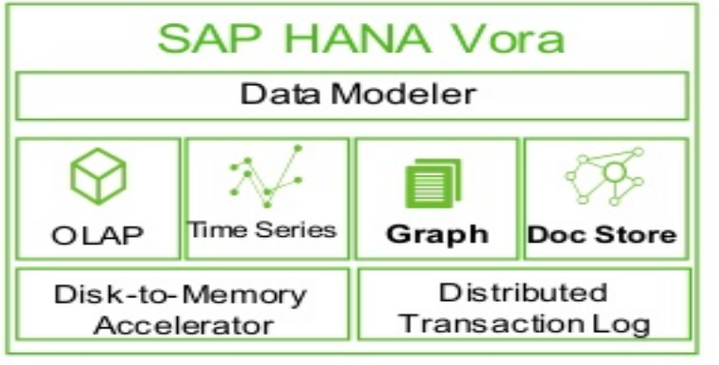
# HANA Vora 1.3 - Relational

- Relational Engines in memory and disk
- In-memory
  - Query compilation
  - Columnar data layout
- Disk based
  - Indices for fast data access



# HANA Vora 1.3 – Graph & Doc Store

- Graph:
  - In-memory and distributed
  - SQL-Like interface for graph analysis
  - Combination of Graph patterns with relational operators
- Doc Store
  - Stored semi structured JSON
  - Compresses in-memory representation
  - Compiled queries with NUMA awareness



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# Summary and Outlook

- Vora allows to combine all data source in an enterprise environment
  - Across different query languages
  - While moving computation close to the data
- Business Insights are driven by all the available data in the enterprise
- Integration into SQL makes it easily consumable

# THANK YOU.

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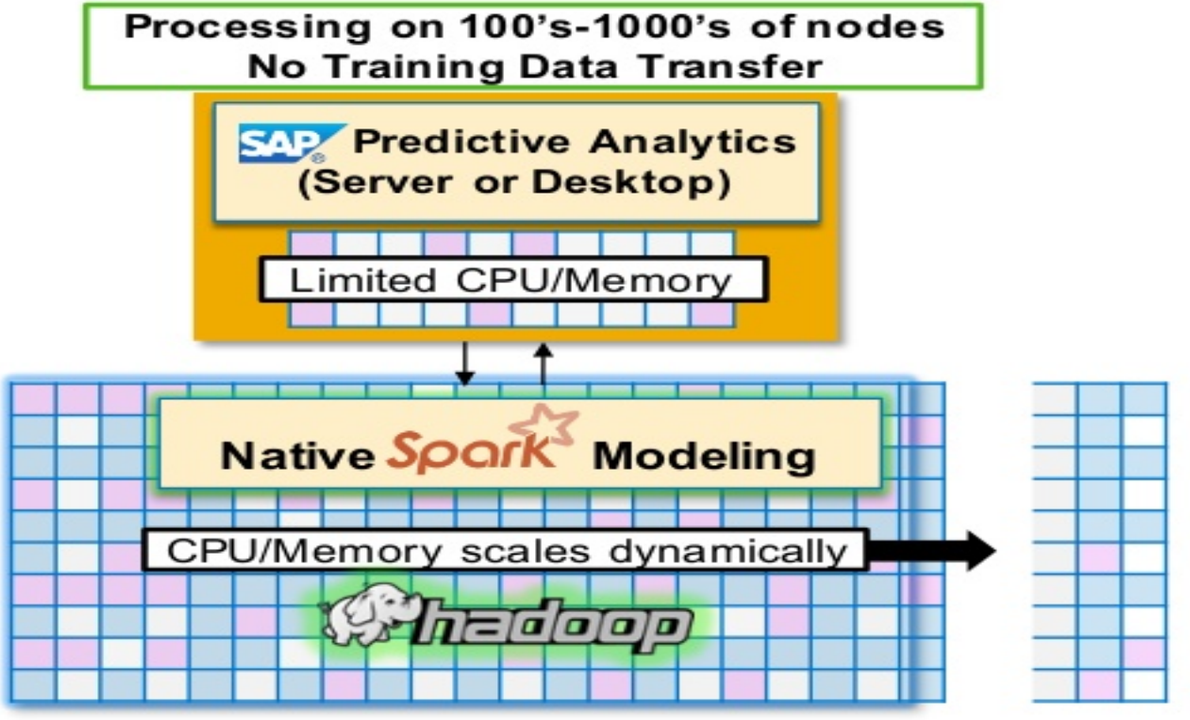
# BACKUP

# HANA Vora 1.3

- Project started 2013 by HANA Research Teams
- Shared concepts and libraries with HANA but independently developed
- Concepts
  - In memory
  - Distributed engines
  - Low memory footprint

SAP HANA Vora

# SAP Predictive Analytics: Optimised for Big Data

- ✓ Native (scala code) Spark approach goes deeper than SQL
- ✓ Performance and Scalability with **Ultra wide datasets**
- ✓ Processing close to the data distributed across the cluster
- ✓ No data transfer